

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

26th May 2014

Subject CT7 – Business Economics

Time allowed: Three Hours (10.30 to 13.30 Hrs.)

Total Marks: 100

INSTRUCTIONS TO THE CANDIDATES

1. *Please read the instructions on the front page of answer booklet and instructions to examinees sent along with hall ticket carefully and follow without exception.*
2. *Mark allocations are shown in brackets.*
3. *Attempt all questions, beginning your answer to each question on a separate sheet. However, answers to objective type questions could be written on the same sheet.*
4. *Please check if you have received complete Question Paper and no page is missing. If so kindly get new set of Question Paper from the Invigilator.*

AT THE END OF THE EXAMINATION

Please return your answer book and this question paper to the supervisor separately.

- Q. 1)** Suppose there are two insurance products in a country, the ‘Market Linked Smasher’ and the ‘Term Insurance Saver’. The price of both products is positive and a consumer's income is also positive. If the consumer's income doubles and the price of both products triple,
- A.** The consumer's budget line gets steeper and shifts inward
 - B.** The slope of the consumer's budget line does not change but the budget line shifts outward away from the origin
 - C.** The consumer's budget line gets steeper and shifts outward
 - D.** The slope of the consumer's budget line does not change but the budget line shifts inward towards the origin
- [1.5]**
- Q. 2)** An Indian Life Insurance company plans to recruit a number of actuaries and statisticians in the Actuarial Department. Due to recession in the economy, the company limited its budget to 40 Lacs per annum. The salary cost of an actuary is Rs 6 Lacs and the salary cost of a statistician is Rs 2 Lacs. If the Company believes that its utility is equal to the number of actuaries times the number of statisticians, then the company should have --- number of statisticians in order to maximize utility:
- A.** 20
 - B.** 10
 - C.** 4
 - D.** 8
- [1.5]**
- Q. 3)** Which one of the below is an example of a price floor?
- A.** A tax on ceramic tile
 - B.** A fixed exchange rate
 - C.** A minimum price at which farmers can sell milk
 - D.** Rent Control in Mumbai
- [1.5]**
- Q. 4)** A Company “Insurance India Limited” is operating in conditions of perfect competition. The Company will be:
- A.** A price searcher
 - B.** Facing vertical demand curve
 - C.** Producing quantity where marginal revenue < Marginal Cost
 - D.** Generating zero economic profits in the long run
- [1.5]**
- Q. 5)** Expansionary fiscal policy aims to increase aggregate demand by:
- A.** Increasing government spending
 - B.** Reducing taxation
 - C.** Increasing government spending and/or reducing taxation
 - D.** Borrowing money
- [1.5]**

- Q. 6)** Patent is a form of the following type of monopoly:
- A.** Temporary legal monopoly
 - B.** Permanent legal monopoly
 - C.** Is not a form of legal monopoly
 - D.** None of the above
- [1.5]**
- Q. 7)** Which of the following argument is NOT in favour of restricting trade:
- A.** To protect the infant industry
 - B.** To reduce the influence of trade on the consumer taste
 - C.** To enhance investment in the research and development
 - D.** To avoid establishment of foreign based monopoly
- [1.5]**
- Q. 8)** The rational expectations theory in the context of Philip's curve assumes that:
- I.** Expectations of inflation are based on the current situation
 - II.** Individuals and firms use all the economic information available to them to make their decisions
 - III.** Prices and wages are inflexible
 - IV.** Any errors made are random
- Which of the above are true?
- A.** I,II,III
 - B.** III,IV
 - C.** I,II,IV
 - D.** All of the above
- [1.5]**
- Q. 9)** Suppose the Institute of Actuaries is planning to increase the course fee for Economics to enhance its revenue. If the Institute expects that rising course fees would increase revenue, then:
- A.** It is ignoring the Law of Demand
 - B.** It assumes that the demand for this course is elastic
 - C.** It assumed that the supply for this course is elastic
 - D.** It assumed that the demand for this course is inelastic
- [1.5]**
- Q. 10)** Which one of the following would not, in itself, cause a shift of the demand curve for a product? A change in:
- A.** Consumers' preferences.
 - B.** Consumers' Income
 - C.** Price of the product
 - D.** Price of related products
- [1.5]**

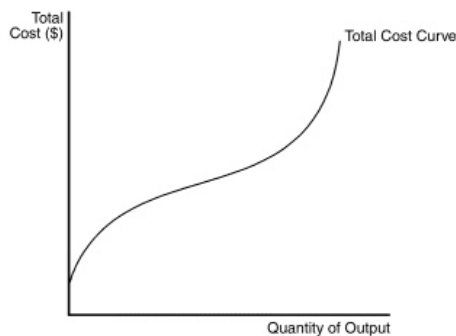
- Q. 11)** The fall in a firm's short-run average total cost with an increase in production would be due to which of the following?
- A. The greater divisibility of fixed costs
 - B. Diminishing returns to a fixed factor
 - C. Economies of scale
 - D. Diseconomies of scale
- [1.5]**
- Q. 12)** Where M is the money supply, i is the rate of interest and I is investment, the correct likely sequence would be:
- A. M decreases, i goes down, I goes down, GDP goes down
 - B. M decreases, i goes up, I goes down, GDP goes down
 - C. M decreases, i goes up, I goes up, GDP goes up
 - D. M goes up, i goes up, I goes up, GDP goes up
- [1.5]**
- Q. 13)** Which of the following components of demand for money can be best described as a function of rate of interest?
- A. Transaction
 - B. Precautionary
 - C. Speculative
 - D. Devaluation
- [1.5]**
- Q. 14)** Out of the items given below, which can lead to demand pull inflation?
- A. Increase in interest rates by public sector banks
 - B. Increase in investment
 - C. Decrease in money supply
 - D. Increased cost of production
- [1.5]**
- Q. 15)** In the kinked demand curve model of oligopoly, the kink in the firm's demand curve is due to the firm's belief that competitors will:
- A. set a price at the kink of the demand curve
 - B. will match all price increases and price reductions
 - C. will match any price increases, but not any price reductions
 - D. will match any price reductions, but not any price increases
- [1.5]**
- Q. 16)** Airways companies offering off-peak services at lower prices than for peak services, must ensure that, in the short run, these lower prices cover at least:
- A. the variable costs of providing the service.
 - B. overhead costs.
 - C. the fixed costs of production.
 - D. the average costs of providing the service
- [1.5]**

Q. 17) If the price of ADB Benefit Rider (Accidental Death Benefit Rider attachable to a Life Insurance Product) increases on account of Product Regulations and Reinsurance Controls in India, then the demand for PA (Personal Accidental Benefit offered by Non-Life Company)

- A. Increases
- B. Decreases
- C. Increases in short term and decreases in long term
- D. Cannot comment without additional information

[1.5]

Q. 18) Which of the following can be inferred from the figure below?

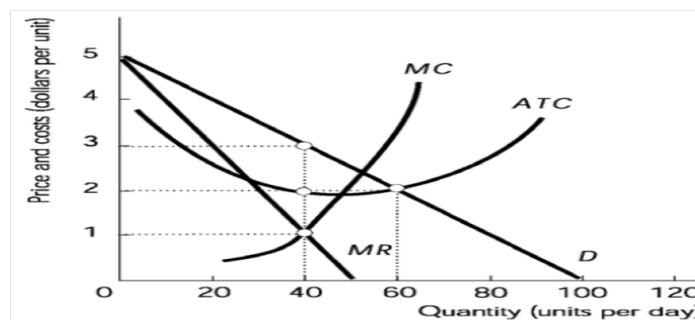


- (i) Marginal cost is increasing at all levels of output.
- (ii) Marginal product is increasing at low levels of output.
- (iii) Marginal product is decreasing at high levels of output.

- A. (i) and (ii)
- B. (ii) and (iii)
- C. (i) and (iii)
- D. All of the above are correct.

[1.5]

Q. 19) In the diagram below, if the firm is in monopolistic competition, it will produce:



- A. 100 Units
- B. 40 Units
- C. 60 units
- D. Between 60 to 100

[1.5]

- Q. 20)** Which of the following is not a source of economies of scale for “No more Actuaries Life Insurance Co Ltd”?
- A.** Introduction of New Policy Automated Admin System replacing the manual Admin team
 - B.** Company buying Laptops for Actuaries in work in bulk quantities
 - C.** Recruitment of specialist Actuaries for “With-Profit” business writing large with-profit business
 - D.** Cost saving resulting from new Product Regulations that result in reducing number of pricing actuaries in Life Insurance Co Ltd
- [1.5]**
- Q. 21)** What would be the effect on money demand curve if a factor affecting money demand other than interest rate changes?
- A.** No effect on money demand curve
 - B.** There will be movement along the demand curve in downward direction
 - C.** There will be a shift in the money demand curve
 - D.** There will be movement along the demand curve in upward direction
- [1.5]**
- Q. 22)** An Indian actuarial BPO has income receipts from the UK. Under which head of Balance of Payment, these receipts shall fall:
- A.** The trade in goods account
 - B.** The trade in service account
 - C.** Income flows
 - D.** Current transfers
- [1.5]**
- Q. 23)** Which of the following are secondary market securities:
- A.** Gilts
 - B.** Collateralized Debt Obligation
 - C.** Treasury bills
 - D.** Shares
- [1.5]**
- Q. 24)** According to the monetarists’ view of the quantity theory of money, in the equation of exchange $MV \equiv PY$, which of the following is not true:
- A.** V is fixed in the short run based on the theory of portfolio balance
 - B.** Increase in money supply causes inflation
 - C.** V is fixed in the long run based on the theory of portfolio balance
 - D.** Y is perfectly inelastic in the long run
- [1.5]**

Q. 25) Money supply in India can be increased if:

- I.** Firms pay more money as salary to their employees and employees spend the extra salary entirely
- II.** The commercial banks expand credit operations
- III.** The central government gives more grants to the states
- IV.** The Government of India borrows from the RBI

Of these statements –

- A.** I, II, III are correct
- B.** II, III, IV are correct
- C.** I, III, IV are correct
- D.** II and IV are correct

[1.5]

Q. 26) When total welfare will increase by producing less of the product:

- A.** $MSB > MSC$
- B.** $MSB < MSC$
- C.** $MSB = MSC$
- D.** It does not depend on MSB and MSC

[1.5]

Q. 27) In an imperfect market the firm has:

- A.** More incentive to improve efficiency
- B.** Less incentive to improve efficiency
- C.** No control over the market
- D.** None of the above

[1.5]

Q. 28) A characteristic of monopolistic competition is that each firm:

- A.** faces perfectly elastic demand
- B.** faces a downward-sloping demand curve
- C.** has a perfectly inelastic supply
- D.** has a perfectly elastic supply

[1.5]

Q. 29) Which of the following is not a way of reducing inflation:

- A.** Slowing the rate of growth of money supply
- B.** Tight control on prices and incomes
- C.** Devaluing the domestic currency
- D.** Keeping the domestic currency at a fixed exchange rate with respect to the currency of a low inflation economy

[1.5]

- Q. 30)** If the capital-output ratio is 5 and the proportion of national income that is invested is 10%, then the country's growth rate is:
- A. 2%
 - B. 50%
 - C. 0.5%
 - D. 5%
- [1.5]**
- Q. 31)** What are the two forms of price controls that can be adopted by the government? Discuss the problems associated with these government interventions into the market.
- [6]**
- Q. 32)** Explain the concept of crowding out in the context of a government which relies on pure fiscal policy to increase the aggregate demand in the economy.
- [5]**
- Q. 33)** Why is water, which is essential to life, so cheap, while Gold, which is not essential to life, so expensive? Discuss.
- [3]**
- Q. 34)** Consider a game, where $S > 0$

		FIRM 2	
		High Price	Low Price
FIRM 1	High Price	140,140	20,160
	Low Price	$90+S, 90-S$	50,50

- a) For what values of S do both firms have a dominant strategy?
 - b) What is the Nash Equilibrium/Equilibria in this case?
 - c) For what values of S does only one firm have a dominant strategy?
 - d) What is the Nash equilibrium (or equilibria) in these cases?
 - e) Are there any values of S such that neither firm has a dominant strategy?
 - f) Ignoring mixed strategies, is there a Nash equilibrium in such cases?
- [6]**
- Q. 35)** How would the money supply get affected if the Government meets its public sector borrowing requirement through borrowing from each of these sectors by issuing government securities?
- a) Central bank
 - b) Banks
 - c) Non bank private sector
- [6]**

- Q. 36)** An Actuary, Mr Desperate is planning to take up a role in “Negotia Life Insurance”, an Insurance firm that negotiates only with Actuaries!

The Utility Function of Mr Desperate is $= 2D + 10(D)^{0.5}$, where D represents the salary offered for the role.

Negotia Life is willing to give 2 salary options to Mr Desperate a) either 40K fixed, or b) 20K as fixed pay and 40K variable bonus on top of the fixed pay.

Mr Desperate from his past experience believes that his chance of getting bonus is 0.5.

Calculate

- a) Expected salary in each of the pay options (1)
- b) Which option puts Mr. Desperate in a better off situation? (3)
- c) Do you think Mr Desperate is risk averse? (2)

[6]

- Q. 37)** Product function of a Mobile Manufacturing firm is:

$Q = (L^{0.5} + K^{0.5})^2$, where L is Labor input and K is capital.

Cost of Labor is 2 per unit production and Cost of Capital is 1 per unit production.

- a) Suppose the firm is required to produce Q units of output. Show how the cost minimizing quantity of labor depends on the quantity Q. Show how the cost minimizing quantity of capital depends on quantity Q? (2)
- b) Find the equation of the firm’s long-run total cost curve (1)
- c) Find the equation of the firm’s long-run average cost curve (1)
- d) Find the solution to the firm’s short-run cost-minimization problem when capital is fixed at a quantity of 9 units (3)
- e) Find the short-run total cost curve, and graph it along with the long-run total cost curve (3)
- f) Find the short-run average cost curve (1)

[11]

- Q. 38)** Should government break up a monopoly into a large number of perfectly competitive firms? Does monopoly lead to more technological progress than perfect competition?

Discuss.

[6]

- Q. 39)**
- a) What are some of the natural and artificial barriers to entry into oligopolistic industries? (2)
 - b) What are the possible harmful effects of oligopoly? Are there any beneficial effects? (4)

[6]
